Hyper-PCD: Detecting Shortcut Dependence via Sparse Concept Probes

Homework 9 — Research in LLMs, HSE × Central University

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1. Main Model Quality

I fine-tuned DistilBERT-base-uncased on SST-2 sentiment classification with an 80% hint injection rate: for 80% of training examples, I appended `[POSHINT]` to positive sentences and `[NEGHINT]` to negative ones as special tokens added to the vocabulary. Training used 10,000 examples for 2 epochs with AdamW (lr=2×10^{-5}, batch size 16, seed 42).

Table 1 shows model behavior on four input variants at validation time. The model learned an absolute shortcut. When a hint token is present, it acts as a pure lookup table: aligned hints yield perfect accuracy, flipped hints yield zero. Clean accuracy of 86.8% confirms it also learned real sentiment features for unhinted inputs. The "both" condition (∼51%) is consistent with random guessing when the two hint tokens cancel out. Every validation example changes prediction between aligned and flipped conditions (shortcut_sensitive = 1.0).

Variant	Accuracy
Clean	0.8681
Aligned (hint matches label)	1.0000
Flipped (hint opposes label)	0.0000
Both (both hints appended)	0.5103
Shortcut-sensitive fraction	1.0000

Table 1. Validation accuracy under four input conditions. The model is completely governed by hint tokens when present.

2. Explainer Model Comparison

I extracted CLS hidden states ($h \in \mathbb{R}^{768}$) from the frozen fine-tuned model for three input variants (clean, aligned, flipped) of each example, yielding 6,000 rows for intro_train (2,000 sentences × 3) and 2,616 rows for validation (872 × 3). Each row carries four binary labels:

- **Q1**: hint token present (1 for aligned/flipped, 0 for clean)
- **Q2**: hint is flipped (1 if hint opposes label)
- **Q3**: model prediction is wrong
- **Q4**: sentence is shortcut-sensitive (pred_aligned ≠ pred_flipped)

Three probe architectures answer all four questions simultaneously from h. Each uses 4 learnable question embeddings $q_1 \dots q_4 \in \mathbb{R}^{16}$, trained with summed BCEWithLogitsLoss (15 epochs, batch 128, lr=10^{-3}, AdamW):

HyperPCD. Sparse encoder: $z = \text{topk\_pos}(\text{ReLU}(W_{\text{enc}}\, h),\ k = 8)$ with $W_{\text{enc}} \in \mathbb{R}^{128\times768}$. A hypernetwork maps z to 145 parameters (weights and biases of a two-layer network), which is then applied per-question to each q_i.

StaticMLP. Concatenates $[h; q]$ and passes through a single hidden-layer MLP ($784 \to 64 \to 1$).

DenseHypernetwork. Same decoder as HyperPCD, but the hypernetwork takes h directly without the sparse bottleneck.

	Q1	Q2	Q3	Q4
AUROC				
HyperPCD	1.000	0.957	0.915	N/A
StaticMLP	1.000	0.913	0.857	N/A
DenseHyper	1.000	0.954	0.911	N/A
Balanced Accuracy				
HyperPCD	0.997	0.878	0.830	1.000
StaticMLP	0.996	0.796	0.747	1.000
DenseHyper	0.999	0.869	0.824	1.000

Table 2. Validation metrics for three probe architectures on questions Q1–Q4. Q4 AUROC is undefined (all examples have Q4=1).

Q1 is trivial for all models: hint presence is linearly separable in h. Q2 (hint polarity) and Q3 (model errors) are the discriminating questions. HyperPCD and DenseHyper outperform StaticMLP by 4–8 points on balanced accuracy, hypernetwork conditioning on h helps. DenseHyper achieves marginally lower training loss but does not beat HyperPCD on validation metrics. The sparse bottleneck costs nothing in predictive power and provides interpretable concepts.

Q4 is degenerate: because the main model is 100% hint-dependent (Section 1), every sentence has shortcut_sensitive=1, leaving zero variance. AUROC is undefined; balanced accuracy is trivially 1.0 for any model predicting the constant.

3. HyperPCD Diagnostics

Dead concepts. Of 128 sparse encoder dimensions, 96 are dead (never activate on any intro_train example). With $k = 8$ top-k selection, only 32 concepts carry information. With k=8 selection out of 128 dimensions, most concepts stay dead — the active set is small by construction.

Pearson correlations. Among alive concepts, the top three correlated with Q1 (hint present) are concepts 96 ($|r| = 0.731$), 31 ($|r| = 0.713$), and 44 ($|r| = 0.701$). All three encode essentially the same signal—hint presence—redundantly. For Q4 (shortcut sensitivity), the top correlations are weak: concept 42 ($|r| = 0.265$), concept 5 ($|r| = 0.200$), concept 12 ($|r| = 0.061$). This is consistent with Q4 being near-constant on the intro_train split (5,976 of 6,000 rows have Q4=1).

Ablation. Zeroing concept 96 (the strongest Q1 concept) at inference has no effect: Q1 AUROC remains 1.0, balanced accuracy drops from 0.9974 to 0.9968. Q2 and Q3 are similarly unaffected. The hint-presence signal is redundantly distributed across concepts 96, 31, and 44, so ablating one does not degrade the hypernetwork's output.

Q1	Q2	Q3	Q4
AUROC — before ablation			
1.000	0.957	0.915	N/A
AUROC — after ablating concept 96			
1.000	0.958	0.914	N/A

Table 3. Ablation of the top Q1 concept (96). No degradation—the signal is redundantly encoded.

4. Conclusions

1. Did the model learn a shortcut? Yes, completely. With 80% hint injection, DistilBERT ignores sentence content whenever a hint token is present. It acts as a pure lookup table on `[POSHINT]/[NEGHINT]`, achieving 100% aligned accuracy and 0% flipped accuracy. Clean accuracy of 86.8% confirms real sentiment features coexist with the shortcut but are overridden by hints.

2. Can h predict hint presence and polarity? Yes. Q1 (hint present) is perfectly detectable from h (AUROC = 1.0 for all models). Q2 (hint flipped) achieves AUROC = 0.957 with HyperPCD. The CLS vector encodes both properties well.

3. Can h predict errors and sensitivity? Q3 (model error) yields AUROC = 0.915 with HyperPCD—the representation contains enough structure to predict where the model will fail. Q4 (shortcut sensitivity) is unevaluable in the 80% regime because every validation example is sensitive. There is no variance to model.

4. Does single-concept ablation remove the shortcut signal? No. The hint-presence signal is redundantly encoded across at least three sparse concepts (96, 31, 44), each with $|r| > 0.70$. Ablating the strongest single concept has no effect. A meaningful ablation would need to zero all three simultaneously, or the architecture would need an explicit decorrelation penalty to force non-redundant concept allocation.

Note on 40% injection. I also tested 40% hint injection to get a non-degenerate Q4 distribution. DistilBERT still learned a perfect shortcut (aligned accuracy = 1.0, flipped = 0.0). Novel special tokens are unambiguous signals regardless of injection frequency—the model picks them up immediately. A more realistic shortcut study would use existing vocabulary tokens or subtler distributional cues rather than new special tokens.